

UNIVERSITY COLLEGE OF ENGINEERING

End Semester Examination Engineering Mathematics-II (BAS203)

Roll No. _____

Subject: Engineering Mathematics-II

Time: 3 Hours

Max. Marks: 70

Code: BAS203

General Instructions:

1. Attempt all questions.
2. The figures in the right-hand margin indicate marks.
3. Use of scientific calculator is permitted.
4. Assume any missing data suitably.

SECTION A — SECTION A

Attempt all questions in this section.

2 marks each | Attempt all questions

Q.No.	Question	Marks	CO
1	A mechanical system is modeled by the differential equation $y'' + 4y = f(x)$. If $f(x) = \sec(2x)$, analyze why the Method of Undetermined Coefficients cannot be applied to find the particular integral and suggest an alternative approach.	2	CO1
2	In a control system circuit, the input voltage is represented by a unit step function $u(t-3)$. Determine the Laplace transform of the rate of change of current, given that the initial current in the circuit is zero.	2	CO2
3	For the power series sum of $[(n!)^2 / (2n)!] * x^n$, evaluate the specific boundary values of 'x' where the D'Alembert's Ratio Test fails to provide a conclusion regarding convergence.	2	CO3
4	A two-dimensional fluid flow is described by a velocity potential $u(x,y)$. If the stream function is $v(x,y) = e^x \sin(y)$, construct the corresponding velocity potential such that the flow remains irrotational.	2	CO4
5	Examine the singularity of the complex function $f(z) = (1 - \cos z) / z^2$ at the origin. Determine if this singularity can be removed and justify your reasoning based on the limit behavior.	2	CO5
6	Consider the variable coefficient equation $x^2 y'' - xy' + y = \ln(x)$. Formulate the substitution required to transform this into a linear differential equation with constant coefficients and state the resulting characteristic equation.	2	CO1
7	A signal $f(t)$ has a Laplace transform $F(s)$. If the signal is time-compressed such that we observe $f(2t)$, demonstrate using the change of scale property how the frequency spectrum is shifted in the s-domain.	2	CO2

SECTION B — SECTION B

Solve all questions. Show detailed steps for full credit.

7 marks each | Attempt all questions

Q.No.	Question	Marks	CO
8	A periodic voltage signal is defined by $f(x) = x^2$ in the interval $(-\pi, \pi)$. By determining the Fourier coefficients, apply Parseval's Identity to find the exact sum of the infinite series $1/1^4 + 1/2^4 + 1/3^4 + \dots$	7	CO3
9	Determine the values of constants 'a' and 'b' such that the function $u = ax^3 + bxy^2$ is harmonic. Once determined, find its harmonic conjugate 'v' and express the complete analytic function $f(z)$ in terms of the complex variable 'z'.	7	CO4
10	Evaluate the complex line integral of $f(z) = \exp(2z) / (z + 1)^4$ along the closed contour C defined by the circle $ z = 2$. Utilize the Cauchy Integral Formula for higher-order derivatives to obtain the result.	7	CO5
11	Two coupled mechanical oscillators are governed by the simultaneous equations: $dx/dt + 2x - 3y = t$ and $dy/dt - 3x + 2y = \exp(2t)$. Solve this system for $x(t)$ and $y(t)$ assuming the system starts from rest at the origin.	7	CO1
12	The transfer function of a signal processing unit is $H(s) = s / (s^2 + a^2)^2$. Employ the Convolution Theorem to find the inverse Laplace transform, effectively determining the unit impulse response of the system in the time domain.	7	CO2

SECTION C — SECTION C

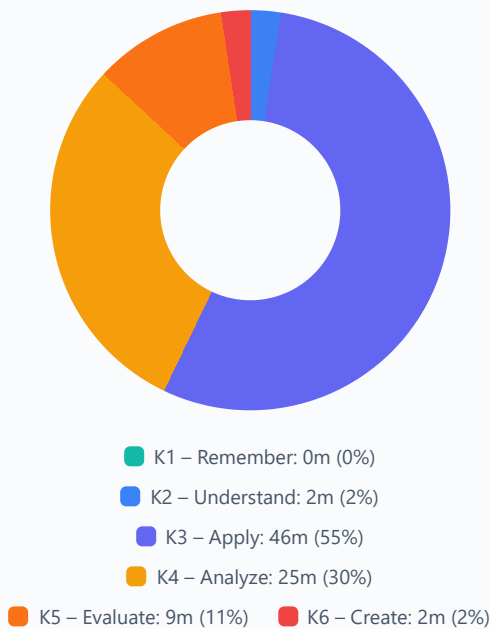
Attempt any ONE part from each question.

7 marks each | Attempt any one part from each question

Q.No.	Question	Marks	CO
13	(a) Investigate the convergence of the infinite series whose n-th term is given by $u_n = [\sqrt{n+1} - \sqrt{n}] / n^p$. Determine the range of values for 'p' for which the series is absolutely convergent.	7	CO3
	<i>OR</i>		
	(b) A heat conducting rod of length $L=1$ is initially at a temperature profile $f(x) = \exp(x)$. To analyze the cooling process, expand this function into a half-range Fourier sine series valid for the interval $(0, 1)$.	7	CO3
14	(a) Find the unique Mobius transformation that maps the points $z_1=0, z_2=1, z_3=\infty$ in the z-plane onto the points $w_1=-1, w_2=-i, w_3=1$ in the w-plane. Verify the transformation for an arbitrary point.	7	CO4
	<i>OR</i>		
	(b) Analyze the conformal mapping defined by $w = z + 1/z$. Identify the critical points where the mapping ceases to be conformal and describe the image of the unit circle $ z =1$ under this transformation.	7	CO4
15	(a) Develop the Laurent series expansion for the function $f(z) = 1 / [(z-1)(z-2)]$ that is valid specifically in the annular region defined by $1 < z < 2$.	7	CO5
	<i>OR</i>		
	(b) Evaluate the real trigonometric integral from 0 to 2π of $[1 / (5 + 4 \cos \theta)] d(\theta)$ by converting it into a contour integral around the unit circle and applying the Residue Theorem.	7	CO5
16	(a) Solve the non-homogeneous differential equation $y'' - 2y' + y = \exp(x) * \ln(x)$ using the Method of Variation of Parameters, ensuring the determination of both the complementary function and the particular integral.	7	CO1
	<i>OR</i>		
	(b) A structural beam deflection is modeled by $(1+x)^2 y'' + (1+x) y' + y = 4 \cos(\ln(1+x))$. Solve this second-order linear differential equation with variable coefficients for the general solution.	7	CO1
17	(a) Derive the Laplace transform of a periodic square wave function $f(t)$ having a period of $2T$, where $f(t) = k$ for $0 < t < T$ and $f(t) = -k$ for $T < t < 2T$.	7	CO2
	<i>OR</i>		
	(b) Use the Laplace transform method to solve the initial value problem $y'' + 9y = \cos(2t)$, given the initial conditions $y(0) = 1$ and $y'(0) = 0$. Explain the physical significance of the resulting terms.	7	CO2

Question Paper Analysis

Bloom's Taxonomy Distribution



Course Outcome Distribution

